8086 & 8088 comparison, and 8087 Coprocessor

Sk. Shalauddin Kabir
Lecturer,
Dept. of Computer Science and Engineering,
Jashore University of Science and Technology.

8086 and 8088 comparison

8086	8088
Similar EU and Instruction set ; dissimilar BIU	
16-bit Data bus lines obtained by demultiplexing $AD_0 – AD_{15}$	8-bit Data bus lines obtained by demultiplexing $AD_0 – AD_7$
20-bit address bus	8-bit address bus
Two banks of memory each of 512 kb	Single memory bank
6-bit instruction queue	4-bit instruction queue
Clock speeds: 5 / 8 / 10 MHz	5 / 8 MHz
	No such signal required, since the data width is only 1-byte

8087 Coprocessor

Co-processor – Intel 8087

Multiprocessor system

- A microprocessor system comprising of two or more processors
- Distributed processing: Entire task is divided in to subtasks

Advantages

- Better system throughput by having more than one processor
- Each processor have a local bus to access local memory or I/O devices so that a greater degree of parallel processing can be achieved
- System structure is more flexible.
  One can easily add or remove modules to change the system configuration without affecting the other modules in the system

Co-processor – Intel 8087

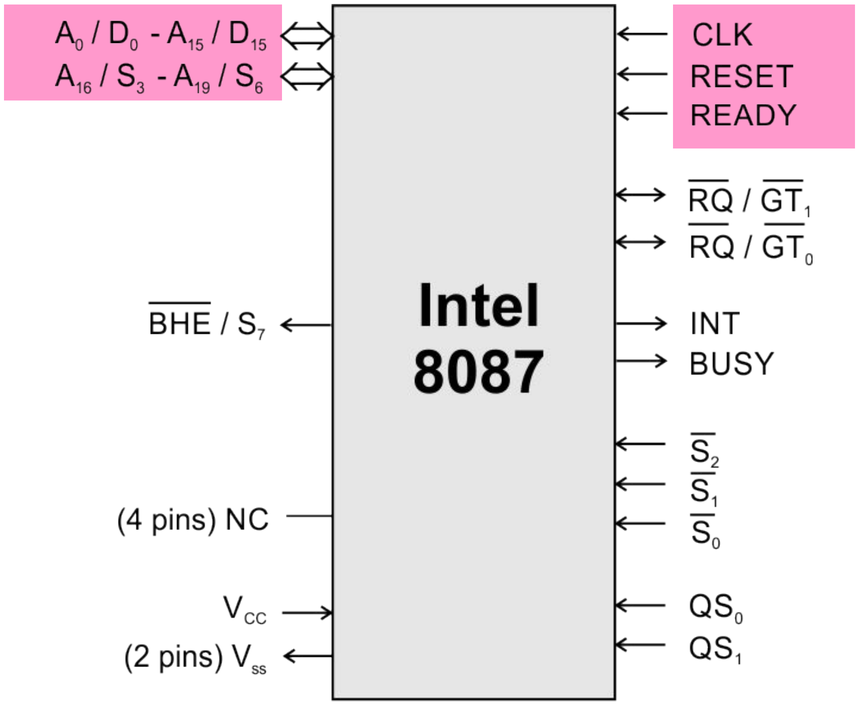
8087 coprocessor

- Specially designed to take care of mathematical calculations involving integer and floating point data
- "Math coprocessor" or "Numeric Data Processor (NDP)"
- Works in parallel with a 8086 in the maximum mode

Features

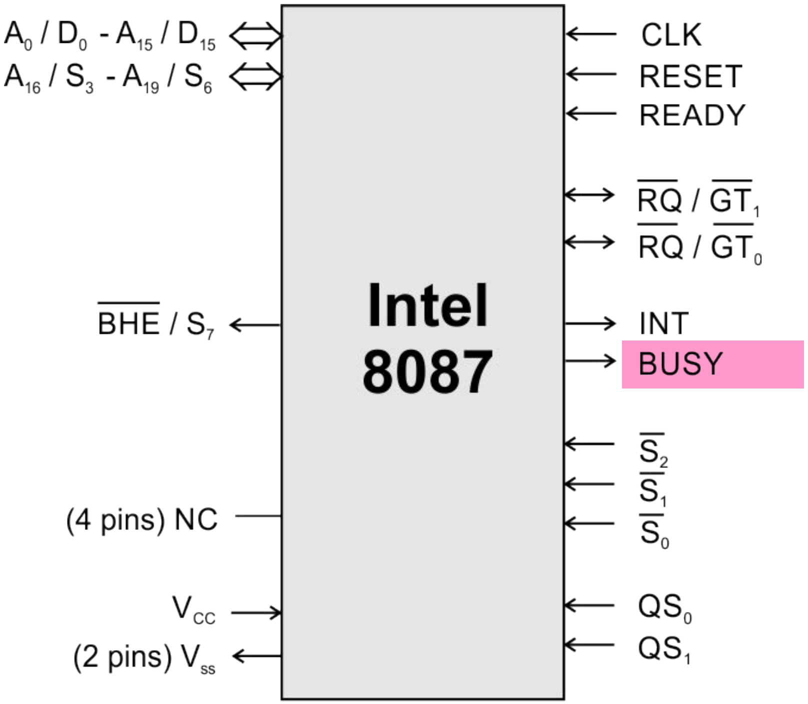
1) Can operate on data of the integer, decimal and real types with lengths ranging from 2 to 10 bytes
2) Instruction set involves square root, exponential, tangent etc. in addition to addition, subtraction, multiplication and division.
3) High performance numeric data processor ⇒ it can multiply two 64-bit real numbers in about 27μs and calculate square root in about 36 μs
4) Follows IEEE floating point standard
5) It is multi bus compatible

Co-processor – Intel 8087



- 16 multiplexed address / data pins and 4 multiplexed address / status pins
- Hence it can have 16-bit external data bus and 20-bit external address bus like 8086
- Processor clock, ready and reset signals are applied as clock, ready and reset signals for coprocessor

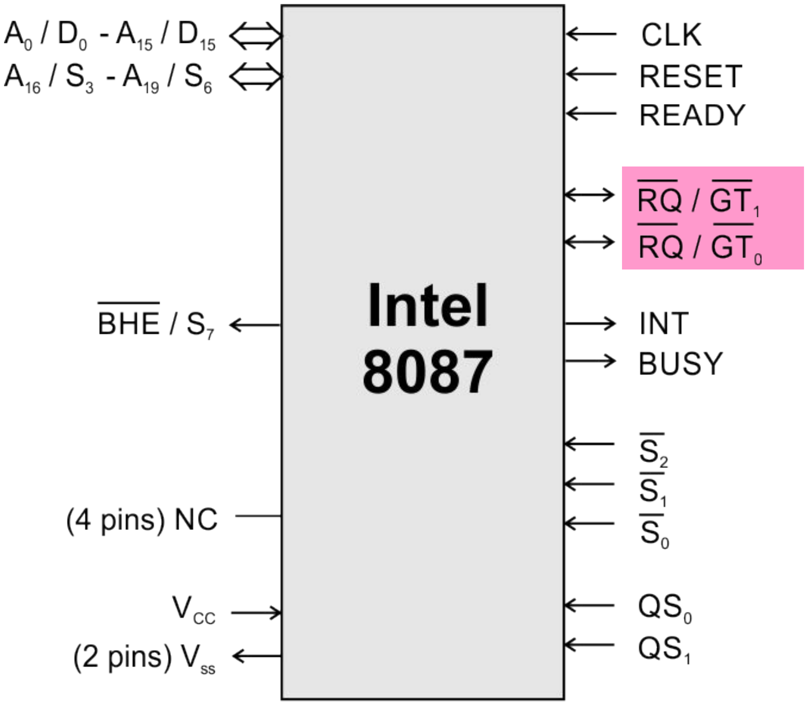
Co-processor – Intel 8087



BUSY

- **BUSY signal from 8087 is connected to the $\overline{\text{TEST}}$ input of 8086**
- **If the 8086 needs the result of some computation that the 8087 is doing before it can execute the next instruction in the program, a user can tell 8086 with a WAIT instruction to keep looking at its $\overline{\text{TEST}}$ pin until it finds the pin low**
- **A low on the BUSY output indicates that the 8087 has completed the computation**

Co-processor – Intel 8087



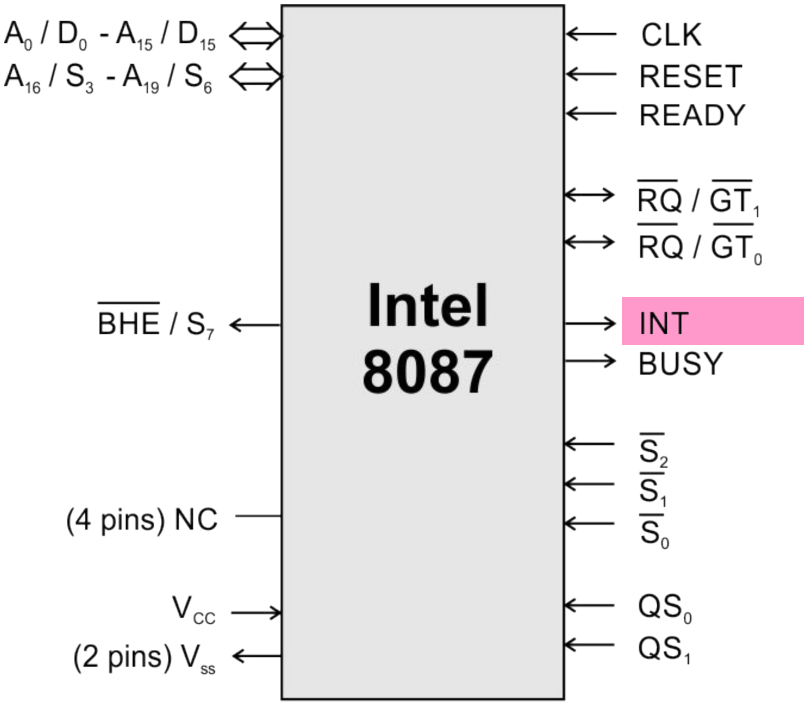
$\overline{RQ} / \overline{GT}_0$

- **The request / grant signal from the 8087 is usually connected to the request / grant ($\overline{RQ} / \overline{GT}_0$ or $\overline{RQ} / \overline{GT}_1$) pin of the 8086**

$\overline{RQ} / \overline{GT}_1$

- **The request / grant signal from the 8087 is usually connected to the request / grant pin of the independent processor such as 8089**

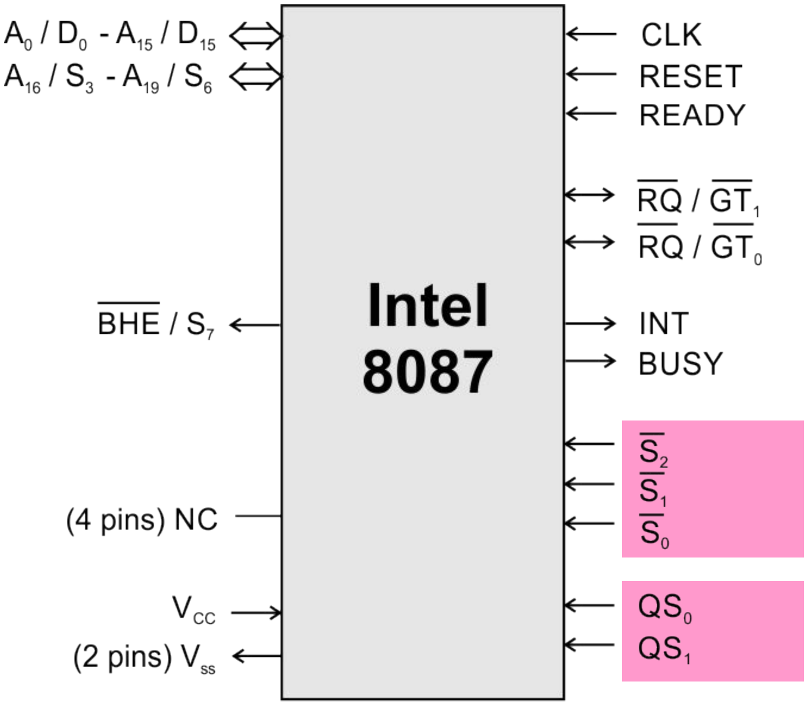
Co-processor – Intel 8087



INT

- **The interrupt pin is connected to the interrupt management logic.**
- **The 8087 can interrupt the 8086 through this interrupt management logic at the time error condition exists**

Co-processor – Intel 8087



$\overline{S}_0$ - $\overline{S}_2$

			Status
1	0	0	Unused
1	0	1	Read memory
1	1	0	Write memory
1	1	1	Passive

QS_0 – QS_1

QS_0	QS_1	Status
0	0	No operation
0	1	First byte of opcode from queue
1	0	Queue empty
1	1	Subsequent byte of opcode from queue

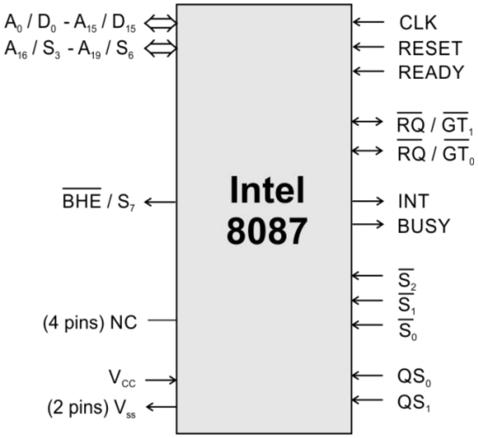
Co-processor – Intel 8087

- 8087 instructions are inserted in the 8086 program

- 8086 and 8087 reads instruction bytes and puts them in the respective queues
- NOP
- 8087 instructions have 11011 as the MSB of their first code byte

- 8087 keeps track for ESC instruction by monitoring $\overline{S_2}$ - $\overline{S_0}$ and AD_0 – AD_{15} of 8086.
- Also keeps track of QS_0 – QS_1.
- Q status 00; does nothing
- Q status 01; 8087 compares the five MSB bits with 11011
- If there is a match, then the ESC instruction is fetched and executed by 8087
- If there is error during decoding an ESC instruction, 8087 sends an interrupt request

- Memory read/ write
- Additional words : $\overline{RQ}$ - $\overline{GT_0}$
- 8087 BUSY pin high
- $\overline{TEST}$
- WAIT



Co-processor – Intel 8087